

Stage 1 — First-pass grounding

👤 User query q

⚙️ BM25 first-pass

Top- $k_1=5$ grounding passages
 $D_1, D_2, \dots, D_5$

Example.

q = “what are the five nouns in Arabic grammar?”

$\Rightarrow D_1, \dots, D_5$: Wikipedia passages on Arabic grammar

Stage 2 — Corpus-grounded LLM expansion

Combine: $q + D_1 \dots D_5$

🧠 Aya 8B LLM

📄 Corpus-grounded
pseudo-document $\hat{d}$

Example. Prompt: “Given the query and passages, write a passage that answers the query in Arabic.”

$\Rightarrow \hat{d}$ = “The five nouns in Arabic are $\bar{a}b\bar{u}$, $\bar{a}kh\bar{u}$, $ham\bar{u}$, $f\bar{u}$, $dh\bar{u}$ (transliterated)...”

Stage 3 — Second-pass retrieval

Concat: $q^\alpha \oplus \hat{d}$ ($\alpha=4$)

⚙️ BM25 second-pass

★ Final top-100 ranking

Example. BM25 input becomes $\underbrace{q \ q \ q \ q}_{\alpha=4 \text{ repeats}} \oplus \hat{d}$,

where α counters term dilution.

$\Rightarrow$ Final nDCG@10 on this query: **1.000** (perfect)